



Explainable Artificial Intelligence: A systematic Review of Progress and Challenges

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ABSTRACT

This work employs a multidisciplinary approach to identify research gaps in the existing literature by presenting a systematic review of systematic reviews on Explainable Artificial Intelligence (XAI). To the best of our knowledge, this is the first thorough meta-review that combines the findings of several excellent reviews to offer a more elevated viewpoint on the goals and difficulties facing the area. The review covers empirical studies published between 2021 and 2023, focusing on high-quality sources. An initial pool of 997 entries was screened across multiple databases, yielding 928 unique articles after duplicate removal. Ultimately, 14 studies met the inclusion criteria and were analyzed in depth. The quality assessment confirmed that all selected reviews adhered to established methodological standards. The key findings show XAI's broad uses, which range from increasing trust and transparency to assisting with financial and management decision-making. The prevalence of healthcare-focused studies emphasizes XAI's importance in enhancing interpretability, fairness, regulatory compliance, and personalized treatment options. Commonly used techniques include visual explanation tools, interpretable machine learning models, and model-agnostic approaches. While the review offers valuable insights, it acknowledges limitations such as its reliance on Q1 journals and the exclusion of broader sources, which may affect comprehensiveness. To advance the field, the study recommends expanding future research to underrepresented domains like autonomous vehicles, defense, and smart cities. It also calls for methodological innovation to enhance accessibility, fairness, privacy, and the development of intuitive explanation strategies. Addressing these gaps can significantly improve the trustworthiness and effectiveness of AI systems across sectors.

1. Introduction

Artificial intelligence (AI) has emerged as a vital force across various sectors, including healthcare, finance, transportation, entertainment, and customer support, due to its capability to perform tasks that mimic human intelligence (Albahri et al., 2023). Utilizing techniques such as machine learning, deep learning, natural language processing, robotics, and computer vision, AI replicates cognitive functions like learning and decision-making. Its effectiveness is largely fueled by big data, which allows for the analysis of patterns and the generation of accurate predictions. Additionally, AI models are categorized into white, grey, and black box models, reflecting different levels of transparency and interpretability in their operations (Ali et al., 2023). White box models provide complete transparency, allowing for a thorough understanding of decision-making. Gray box models give partial transparency, striking a

balance between complexity and clarity. Black box models, notably in DL, perform complicated tasks with great precision but lack transparency, which raises issues in instances when understanding decision rationales is crucial (Ali et al., 2023) (see Fig. 1).

In pattern recognition applications such as picture classification, object detection, and time series analysis, XAI is crucial for increasing the interpretability of complex models such as deep neural networks. Methods such as Class Activation Mapping (CAM) and Grad-CAM provide visual explanations to assist users understand what elements a model concentrates on during prediction, thereby increasing user trust and providing diagnostic insight, particularly in medical imaging and autonomous driving. Highlighting XAI's involvement in these activities emphasizes its importance to pattern recognition academics and practitioners.

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2. Background

2.1. What is XAI?

It seems difficult to determine when the term Explainable Artificial Intelligence was first coined, but in 2016 DARPA (Defense Advanced Research Projects Agency) issued its encouraging and regulatory announcement DARPA-BAA-16-53¹ for this field (Barredo Arrieta et al., 2020). DARPA also defined XAI as AI systems that articulate their reasoning to users, highlighting both the advantages and disadvantages for forecasting future actions 11. The objective is to equip human experts with insights into the reasons behind AI's decisions. In another word, XAI is the attempt to give answer to "Why?" question (Chazette & Schneider, 2020). Furthermore, the General Data Protection Regulation (GDPR) serves as a significant illustration of Europe's increasing requirements for XAI. The GDPR enforces data protection and privacy standards for all residents and entities within the European Union (EU) that use personal information in automated decision-making processes. Under its directives, organizations are obligated to reveal the process by which AI algorithms make final decisions, enabling users to identify any possible biases (Chen et al., 2022).

2.1.1. XAI stages

XAI process can go through three stages:

- Explainable Building Process
- Explainable Decision
- Explainable Decision Process

XAI development takes three stages. The first stage focuses on leveraging visualization tools and simulations to improve comprehension for non-technical users, hence fostering trust and interaction. The second step focuses on providing understandable explanations for AI judgments, using techniques such as SHAP and LIME, which are critical in supervised systems like financial fraud detection. The last stage incorporates AI into broader business processes, allowing for seamless interactions between AI outputs and business rules, hence facilitating sophisticated applications such as tailored customer engagement. See Fig. 2: Three Stages of XAI Development: From Visualization to Business Integration

2.1.2. XAI goals

As shown in Fig. 3, XAI goals are divided into nine goals: trustworthiness, causality, transferability, informativeness, confidence, fairness,

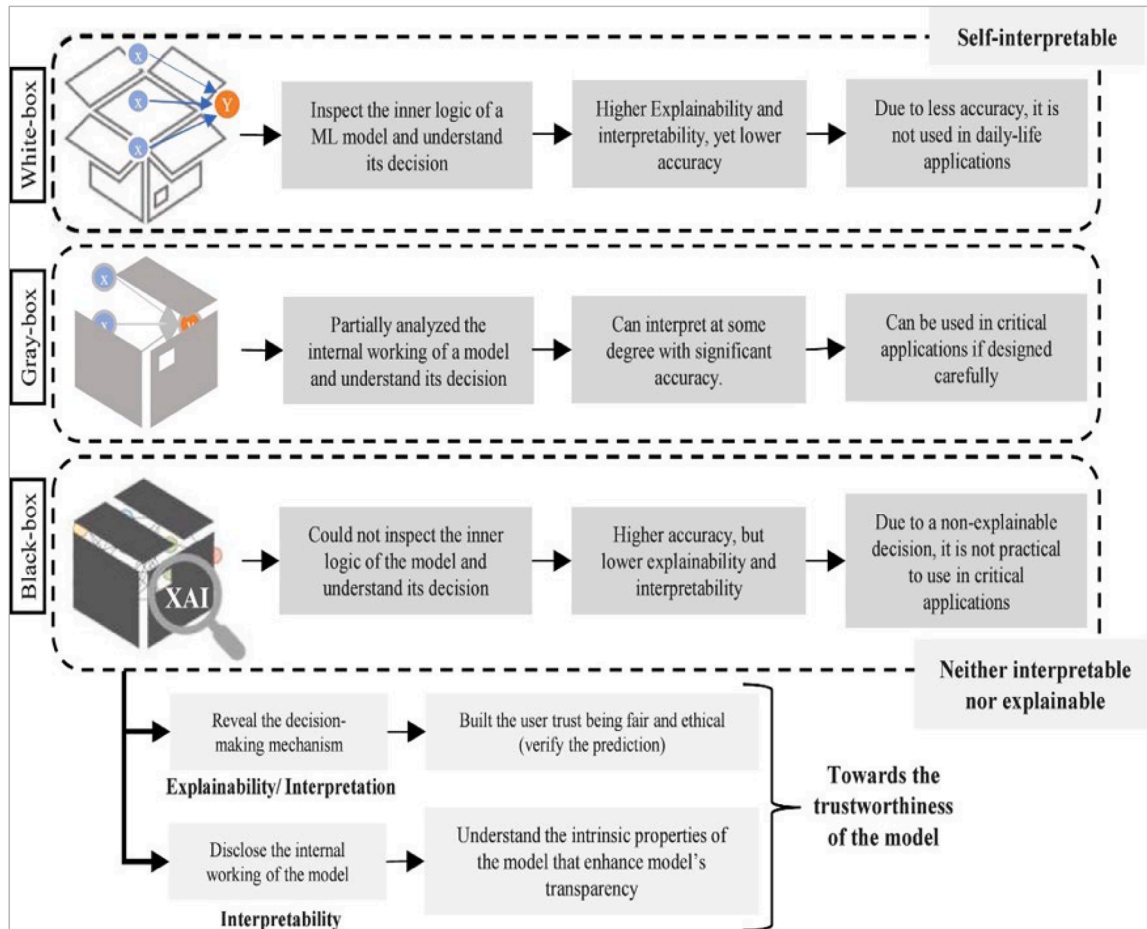


Fig. 1. Illustration of white, gray, and black box models, highlighting their transparency levels in AI systems (Source: Ali et al., 2023). The black box dilemma in AI presents challenges for developing trust and security in AI systems. This issue has given rise to XAI, which seeks to balance interpretability and accuracy using white or gray box models while also providing explanations for highly accurate but opaque black-box models to maintain transparency in decision-making (Alsaleh et al., 2023).

accessibility, interactivity, and privacy awareness (Díaz-Rodríguez et al., 2022).

¹ <https://www.darpa.mil/attachments/DARPA-BAA-16-53.pdf>

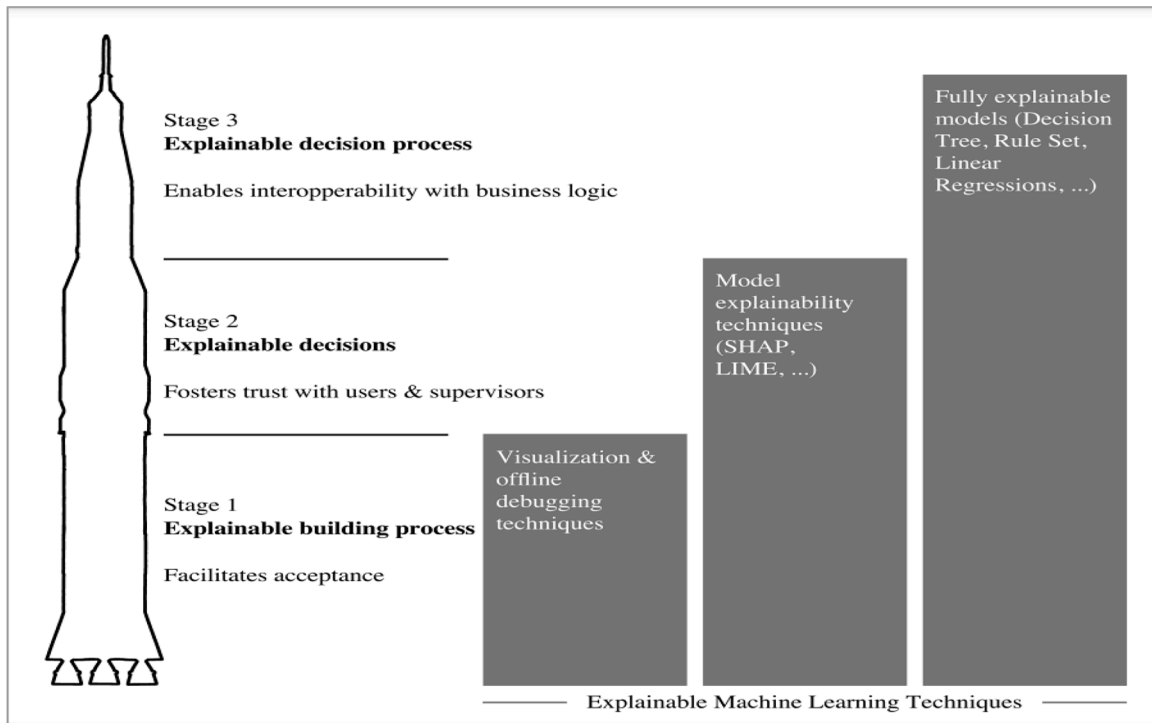


Fig. 2. Three Stages of XAI Development: From Visualization to Business Integration.

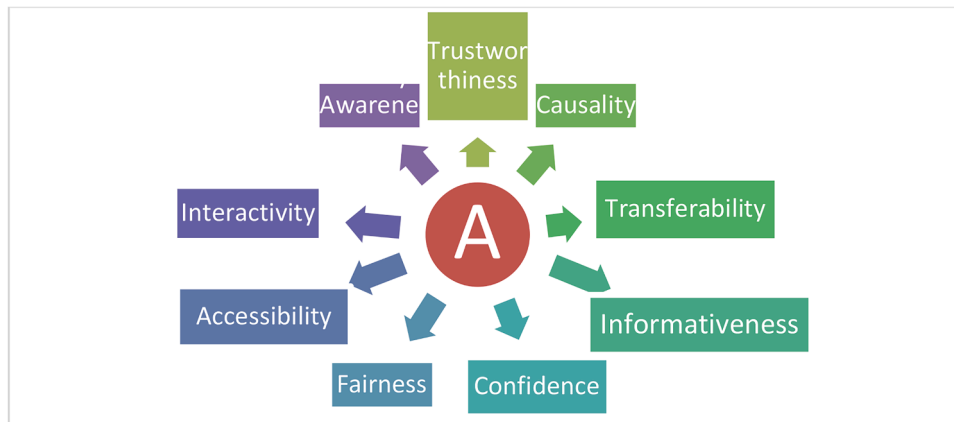


Fig. 3. Key goals of Explainable Artificial Intelligence (XAI).

1. **Causality:** Identifying causal relationships between variables is crucial in XAI. It requires prior knowledge to understand these connections, mainly targeting domain experts (Giuste et al., 2023).
2. **Transferability:** Focuses on applying knowledge from one domain to another. XAI clarifies model limitations and applicability, though not all models ensure easy transferability (Groen et al., 2022).
3. **Informativeness:** An explainable model provides key information for user decision-making, aiming to clear any misconceptions by revealing the model's internal workings (Samek et al., 2021).
4. **Confidence:** A model is considered confident if it reliably performs its tasks, offering users a sense of stability and trust in its outputs (Hauser et al., 2022).
5. **Fairness:** Ensures ethical use of AI, requiring models to transparently show their decision-making processes to prevent bias and unfair practices (Holzinger et al., 2022).
6. **Accessibility:** Advocates for user-friendly interfaces in XAI, suggesting visual explanations to make the model's inner workings accessible to users (Huang et al., 2022).
7. **Interactivity:** Emphasizes the need for user-friendly interactions between AI systems and users, ensuring that information is easily understandable (Jung et al., 2023).
8. **Privacy Awareness:** Important in XAI, it involves safeguarding user data, highlighting the need to consider privacy throughout the system development lifecycle (Linardatos et al., 2020).
9. Recent improvements in XAI for time series and human motion analysis have given insight on the techniques' larger uses. (Vielhaben et al., 2024) uses virtual inspection layers in time series analysis to improve sequential data transparency. Furthermore, (Olivas-Padilla et al., 2024) presents a complete framework for analyzing and producing human motion data using XAI, increasing its applicability in mimicking complicated human interactions. Recent advancements have further expanded the use of XAI techniques to time series data in fields including financial forecasting, healthcare (e.g., ECG analysis), and industrial monitoring. In order to successfully comprehend temporal patterns and anomalies, practitioners have used techniques such as attention-based models and modified gradient-based

explanations to interpret sequential data. These findings support (Yu et al., 2022), which presents eX-ViT, an explainable vision transformer that addresses explainability difficulties in weakly supervised semantic segmentation. Furthermore, (Bai et al., 2021) provides an essential addition by offering an explainable neural network specifically designed for Alzheimer's disease detection, demonstrating the expanding importance of XAI in medical applications. (Yu et al., 2023) provide a comprehensive survey of recent advances in explainable deep learning for robust and efficient pattern recognition, emphasizing the necessity of transparency in these systems.

2.1.3. XAI methods

Explaining AI model predictions is critical for promoting transparency and confidence across businesses. XAI techniques improve machine learning models' interpretability, especially in healthcare, finance, and law. These methods simplify complex forecasts, emphasize key aspects, and examine decision-making processes, making it easier to debug, choose features, and identify biases. Finally, they improve the accountability and usefulness of AI systems in key decision-making scenarios (see Table 1).

Table 1
XAI methods and applications.

No.	Method	Description & Applications
1	Anchors	Provides rule-based explanations for predictions, aiding decision-making in legal and financial sectors by identifying stable prediction conditions.
2	LRP (Layer-wise Relevance Propagation)	Backpropagates relevance through neural networks to explain feature contributions, especially in image recognition and healthcare.
3	Deep Taylor Decomposition	Uses Taylor expansion for visualizing feature importance in image classification, useful in computer vision and remote sensing.
4	Prediction Difference Analysis	Evaluates model sensitivity by hiding features, aiding in feature selection across finance, e-commerce, and healthcare.
5	XGNN	Explains graph neural network outputs by identifying representative graphs; applied in network security and chemical analysis.
6	SHAP	Uses game theory to assign feature importance for both local and global interpretation, widely used in finance and healthcare.
7	Asymmetric Shapley Values	Adds causal insights to SHAP, useful in epidemiology and economics for understanding intervention impacts.
8	Break-Down	Provides additive explanations of feature contributions, beneficial in finance, marketing, and sports analytics.
9	Shapley Flow	Enhances complex model interpretability by distributing importance via a causal graph.
10	Textual Explanations of Visual Models	Converts image features into descriptive text, improving media accessibility and tools for visually impaired users.
11	Integrated Gradients	Calculates gradients to explain model predictions, aiding in feature attribution for image and text analysis.
12	Causal Models	Uses counterfactuals and causal structures for policy and healthcare decision-making.
13	Meaningful Perturbations	Detects key features by analyzing prediction shifts from input changes, crucial in cybersecurity and adversarial defense.
14	X-NeSyL (Explainable Neural-Symbolic Learning)	Merges symbolic reasoning with neural learning for interpretable AI, applied in robotics and logic tasks.
15	CAM (Class Activation Mapping)	Highlights image regions influencing predictions using CNNs, aiding diagnostics in medical imaging.
16	Grad-CAM	Uses gradients to visualize key prediction areas in CNNs, applied in healthcare, autonomous driving, and visual tasks.

1) Recent Advances in AI for Industrial Diagnostics and Object Detection

Recent AI advances for industrial applications have made tremendous progress in addressing complicated diagnostic and detection challenges. For example, (Tao et al., 2023) proposed a Planetary Gearbox Fault Diagnosis technique based on FDKNN-DGAT that effectively detects faults with little labeled data, addressing a common real-world restriction. Similarly, (Tao et al., 2024) introduced an Enhanced Feature Extraction YOLO technique designed for small object recognition in industrial contexts; their model achieves high precision in cluttered scenes by incorporating receptive-field attention and multi-scale characterist. Furthermore, (Song et al., 2024) developed a Saturated-Threshold Event-Triggered Adaptive Global Prescribed Performance Control method for nonlinear Markov jump systems, which has been applied to chemical reactor control, demonstrating the potential of AI in complex, dynamic systems. While these models push the limits of AI capabilities in industrial contexts, interpretability remains critical. Integrating explainability techniques into such advanced models can improve transparency, increase user trust, and promote greater acceptance in safety-critical fields.

2) XAI for Pattern Recognition

Modern AI applications in industrial diagnostics rely heavily on pattern recognition tasks such as picture categorization, object detection, and neuroimaging analysis. Grad-CAM, Layer-wise Relevance Propagation (LRP), and other visual explanation approaches help us understand model decision-making by emphasizing key aspects in input data. These explainability approaches are especially useful in high-risk industries like as healthcare and manufacturing, where interpretability influences trust and regulatory compliance. Incorporating XAI into pattern recognition improves model transparency, simplifies error analysis, and allows for more resilient deployment in real-world applications.

In addition to image-based techniques, explainability methods for time series data are gaining popularity because of their utility in applications such as predictive maintenance, financial forecasting, and patient monitoring. Virtual inspection layers, for example, aid in the visualization of prominent regions across temporal sequences, improving interpretability in sequential decision making. Furthermore, feature attribution tools such as SHAP and LIME have been modified for time series classification and early warning systems, allowing practitioners to better understand which time-dependent variables have the most impact on a model's prediction. This expansion into temporal domains promotes a broader and more inclusive understanding of pattern recognition in XAI research.

3) Visual Explanation Techniques for Interpretability in Pattern Recognition

Visual explanation methods, such as Layer-wise Relevance Propagation (LRP), are essential for comprehending the decision-making processes of deep learning models, particularly convolutional neural networks. These algorithms generate heatmaps that highlight salient regions within input photos, allowing users to determine which sections of the data contribute the most significantly to the model's predictions.

Grad-CAM, in particular, has received attention due to its ability to generate class-discriminative localization maps without altering the underlying model architecture. It generates visual overlays that identify critical regions that influence classification outcomes by computing the gradients of a target class in relation to feature maps.

This method is especially effective in high-risk fields such as industrial diagnostics, medical imaging, and object identification, where interpretability directly affects trust, regulatory compliance, and error analysis. Implementing these visual explanation techniques in industrial

pattern recognition tasks enhances transparency by providing understandable visual explanations that help stakeholders understand, identify faults, and accept models. Their compatibility with numerous CNN designs, as well as their low computing requirements, making them versatile tools for embedding explainability into advanced artificial intelligence systems. Furthermore, integrating these methodologies with others, such as LRP, may yield complementary insights, expanding the interpretability landscape in complex, safety-critical systems.

In conclusion, visual explanation approaches like these are critical tools for bridging the gap between high-performing AI models and human interpretability, generating more trust and promoting safe deployment in industrial settings. Their incorporation into pattern recognition processes highlights transparency as an important aspect of modern AI development.

2.2. About the study

The challenge faced by this study is that the topic of XAI is widely discussed among researchers. Consequently, systematic reviews of it are very recent, making it easy for researchers to find reviews published in the years 2022 and 2023. The time gap that needs to be filled with recent reviews is very narrow. Hence, the initiative for this study is to conduct a systematic review of the available systematic reviews, even the highest-rated ones, by requiring only reviews published in journals classified as Q1. This choice is encouraged by the scarcity of systematic reviews of systematic reviews in XAI, which accounts for this study being almost the first systematic review of systematic reviews in XAI. This review presents the following five research questions as its focus:

- RQ1: What domains are covered by XAI studies?
- RQ2: What goals do XAI studies aim to achieve?
- RQ3: What are the main methods used in XAI research?
- RQ4: What gaps exist in current XAI research?
- RQ5: What recommendations are made within the XAI field?

The structure of this systematic review is organized as follows: **3. Methodology:** This section outlines the methodology used to select and review the XAI systematic reviews. **4.Results and Findings** Answers to the five research questions based on the selected studies. **5.Discussion:** This section discusses the results of the review. **6. Conclusion:** Summarizes the findings, highlights limitations, and provides recommendations for improvement and future exploration. **7. Future Work:** Outlines potential areas for further investigation based on the study's outcomes and identified gaps, suggesting how future work can extend or

enhance the current research. [36].

3. Methodology

Reviewing past literature is a critical component of all scholarly work. Conducting a thorough review lays the groundwork for advancing knowledge. It aids in theory formulation, addresses areas with extensive research, and highlights where additional investigations are needed (T et al., 2022). Accordingly, this systematic review of systematic reviews is conducted to bolster XAI. The methodology of this review unfolded in stages: setting inclusion and exclusion parameters for the studies gathered, determining data sources, and devising a search strategy in line with PRISMA guidelines, and evaluating the quality of the studies.

3.1. Inclusion and exclusion criteria

The decision to accept or reject the paper for this review is outlined in Fig. 4, which details the inclusion and exclusion criteria.

3.2. Data source and search strategy

This systematic review was undertaken by searching databases such as ScienceDirect, Sage, IEEE Xplore, Springer, Taylor & Francis, PubMed, ACM, and Google Scholar, focusing on publications with titles that included terms such as "Explainable AI" or "XAI" as well as "systematic review." The Database Selection Heatmap (Fig. 5) visualizes the distribution of the 622 articles Following PRISMA rules (see Fig. 6) and exclusion criteria, 608 papers were deleted, leaving only 14 systematic review studies for analysis. Also, Fig. 6 presents data from these selected works, emphasizing their scholarly effect, with citation counts ranging from 0 to 223. These reviews cover 997 publications, which have been reduced to 928 unique research after removing duplicates. “

The evaluated articles, which range from 2005 to 2023, provide insights into the evolution of XAI over time. This vast temporal range emphasizes both historical context and modern advances in the subject. The studies encompass a variety of sub-domains, demonstrating XAI's versatility and depth across industries, such as improving trustworthiness in healthcare and helping management decision-making. Fig. 7 illustrates the citation frequency of each selected systematic review, reflecting their academic impact and relevance within the XAI research community. Table 2 Selected systematic review studies on XAI: details include article identifiers, publication year, citation counts, reviewed articles, research domains and subdomains, data sources, journals, ISSN, and Journal SJR quartiles.

Inclusion	Exclusion
• The study should be a systematic review of the use of Explainable Artificial Intelligence in specific domain(s). • The study should be published in Q1 journal as per SJR ranking. • Accessible as per the existing BUID library database, and/or open source. • Published after 1/1/2019. • Written in English.	• The study is survey, not a systematic review. • The study focuses on XAI methods, features, goals, and other aspects without being tied to a specific domain such as healthcare or finance. • Published in Q2 journal or lower. • The study is a conference paper, book chapter, Ph.D, thesis, master dissertation or preprint (e.g, published only in arXiv or ResearchGate)

Fig. 4. Inclusion and exclusion criteria.

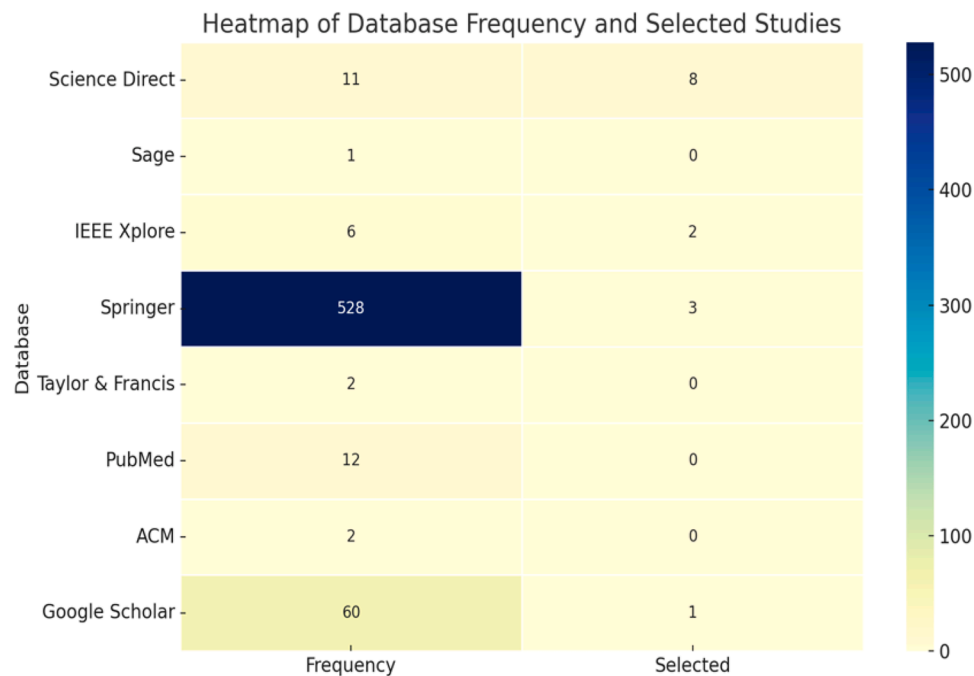


Fig. 5. XAI methods and their application.

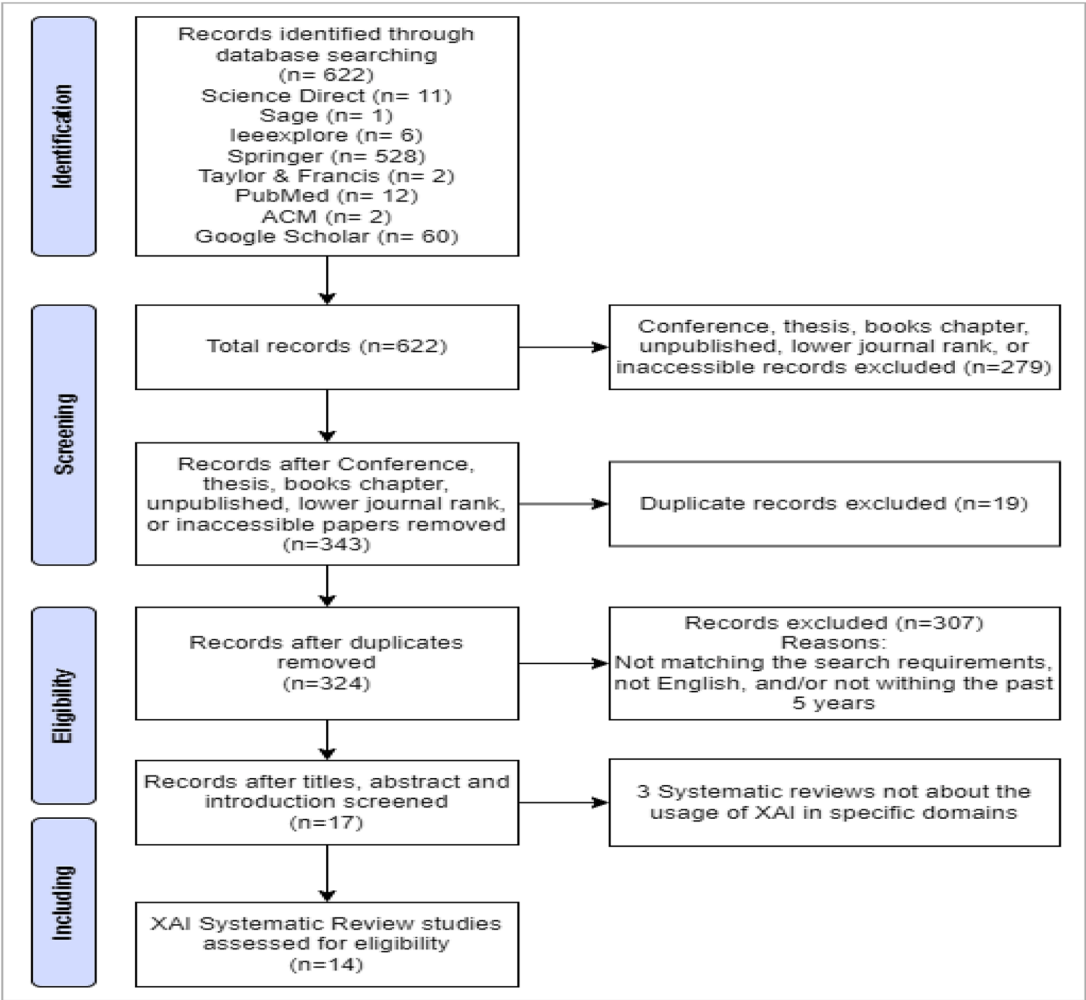


Fig. 6. PRISMA flowchart of the selected systematic reviews.

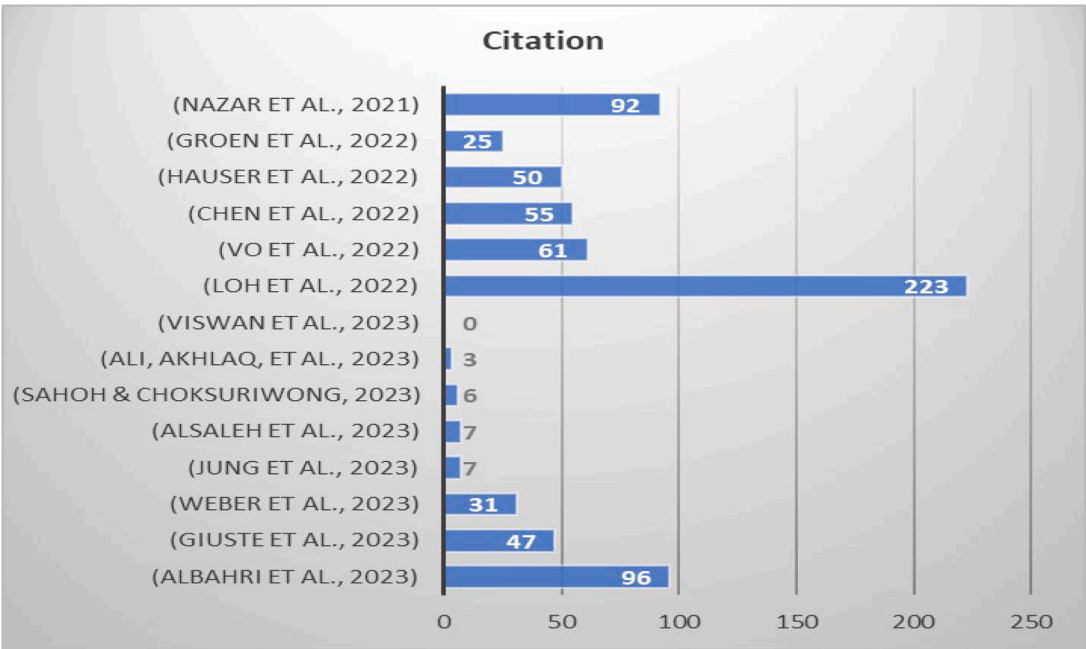


Fig. 7. Citation of the selected systematic reviews.

Fig. 8 shows the annual publication trend of selected systematic reviews from 2019 to 2023, highlighting the growing scholarly interest in Explainable Artificial Intelligence. Fig. 9 presents the number of primary studies included in each systematic review, indicating the depth and breadth of evidence synthesis in each study.

The two years following the reference name represent the timespan of the reviewed articles in the specified study, indicating the oldest article and the newest one. For example, (T et al., 2022): 2018–2022 indicates that this systematic review reviewed articles published between 2018 and 2022.

3.2.1. Quality assessment

A checklist of seven criteria was created to evaluate the quality of the selected studies. The checklist was not intended to criticize individual researchers, but rather to assess alignment with the research objectives. Each study was assessed based on the following responses: "Yes" (1 point), "No" (0 points), and "Partially" (0.5 points). The total score for each study ranged from 0 to 7, with higher scores indicating a better fit with the research aims (see Table 3).

4. Results and findings

This section presents the outcomes of this systematic review, organized according to five research questions. It involves an analysis of 14 systematic reviews on XAI published between 2021 and 2023.

4.1. RQ1: XAI covered domains

The research finds that these studies cover a wide range of domains, with a strong focus on healthcare, followed by specialized contributions to finance and management. The synthesis of these studies in response to RQ1 reveals a varied landscape in which XAI is being used to address challenging challenges across multiple sectors.

This field is the most intensively investigated in XAI, with numerous studies looking into its sub-domains. For example, (Albahri et al., 2023) examines XAI's involvement in diagnostic systems, emphasizing its importance in increasing openness for medical professionals. The study (Alsaleh et al., 2023) underlines the potential of XAI in clinical decision assistance to improve patient outcomes. Research (Barredo Arrieta et al.,

2020) looks into how it might be used to analyze electronic health records while also addressing data privacy and patient care. Furthermore, (Groen et al., 2022) studies XAI's use in interpreting imaging techniques for disorders such as cancer, while (Vielhaben et al., 2024) discusses ethical considerations. Studies (Olivas-Padilla et al., 2024) and (Bai et al., 2021) look at the obstacles of using XAI in clinical settings and its impact on decision-making, while (Madumal et al., 2020) discusses its function in disease prediction and management. The study (Pinto dos Santos & Baeßler, 2018) underlines the importance of XAI in ensuring fairness and accountability in AI systems in health settings. Furthermore, (Shin, 2021) underlines the need of interdisciplinary collaboration in developing effective XAI solutions and provides a comprehensive overview of XAI's applications in a wide range of health-related settings. Vo et al. (2022) demonstrate novel techniques to incorporating XAI into healthcare, highlighting its transformative potential.

Diagnostics and illness prediction are a common issue in XAI research, with (Albahri et al., 2023) demonstrating its application in identifying disorders such as glaucoma and tracking Alzheimer's disease progression, highlighting XAI's importance in increasing diagnostic accuracy.

Medical Imaging is Several studies have demonstrated the effectiveness of XAI in understanding medical pictures from X-rays, CT scans, MRIs, and ultrasounds for disorders such as COVID-19 and breast cancer. Study (Albahri et al., 2023) reveals how XAI improves the interpretability of complicated imaging data, hence increasing diagnostic accuracy and physician confidence. The role of XAI in detecting early illness indicators and providing timely therapies is investigated in the study (Alsaleh et al., 2023). Reference (Groen et al., 2022) demonstrates how visual explanation techniques explain AI decision-making, making it more understandable to healthcare practitioners. Study (Olivas-Padilla et al., 2024) addresses issues such as transparency and bias in AI algorithms, whereas study (Pinto dos Santos & Baeßler, 2018) stresses XAI's role in guaranteeing fairness and accountability in medical imaging. Together, these findings demonstrate XAI's transformative potential for increasing the effectiveness and reliability of medical imaging systems.

Research (Alsaleh et al., 2023; Madumal et al., 2020; Vielhaben et al., 2024), and (Pinto dos Santos & Baeßler, 2018) emphasize XAI's significance in improving Clinical Decision Support (CDS) systems. A

Table 2
Selected systematic review details.

Article Code	Article Reference	Year	Cited by	# Reviewed articles = 997	# Reviewed articles without duplication = 928	Year Range of Reviewed Articles	XAI Domain	XAI Subdomain	Database	Journal	ISSN	SJR
S1	(Albahri et al., 2023)	2023	96	64	51	2012–2023	Healthcare	Trustworthiness	Science Direct	Information Fusion	1566–2535	Q1
S2	(Groen et al., 2022)	2022	25	179	179	2016–2021	Healthcare	Radiology	Science Direct	European Journal of Radiology	0720–048X	Q1
S3	(Loh et al., 2022)	2022	223	97	89	2018–2022	Healthcare	Applications	Science Direct	Computer Methods and Programs in Biomedicine	0169–2607	Q1
S4	(Jung et al., 2023)	2023	7	6	4	2020–2021	Healthcare	XAI properties	Science Direct	Heliyon	2405–8440	Q1
S5	(Hauser et al., 2022)	2022	50	37	34	2017–2021	Healthcare	Skin cancer	Science Direct	European Journal of Cancer	0959–8049	Q1
S6	(Vo et al., 2022)	2022	61	94	93	2005–2021	Healthcare	Drug-drug interactions	Science Direct	Computational and Structural Biotechnology Journal	2001–0370	Q1
S7	(Alsaleh et al., 2023)	2023	7	22	22	2009–2023	Healthcare	Prediction of disease comorbidity	Science Direct	International Journal of Medical Informatics	1386–5056	Q1
S8	(Ali, Akhlaq et al., 2023)	2023	3	92	89	2019–2022	Healthcare	General	Science Direct	Computers in Biology and Medicine	0010–4825	Q1
S9	(Giuste et al., 2023)	2023	47	93	80	2010–2022	Healthcare	Pandemics	IEEE Xplore	IEEE Reviews in Biomedical Engineering	1941–1189	Q1
S10	(Nazar et al., 2021)	2021	92	88	85	2016–2021	Healthcare	Human Computer Orientation	IEEE Xplore	IEEE Access	2169–3536	Q1
S11	(Chen et al., 2022)	2022	55	68	54	2012–2021	Healthcare	Medical Imaging	Nature	NPJ digital medicine	2398–6352	Q1
S12	(Viswan et al., 2023)	2023	0	37	30	2018–2023	Healthcare	Alzheimer	Springer	Cognitive Computation	1866–9964	Q1
S13	(Weber et al., 2023)	2023	31	60	59	2010–2021	Finance	Applications of XAI	Springer	Management Review Quarterly	2198–1639	Q1
S14	(Sahoh & Choksuriwong, 2023)	2023	6	60	59	2010–2020	Management	Decision Making	Springer	Journal of Ambient Intelligence and Humanized Computing	1868–5145	Q1

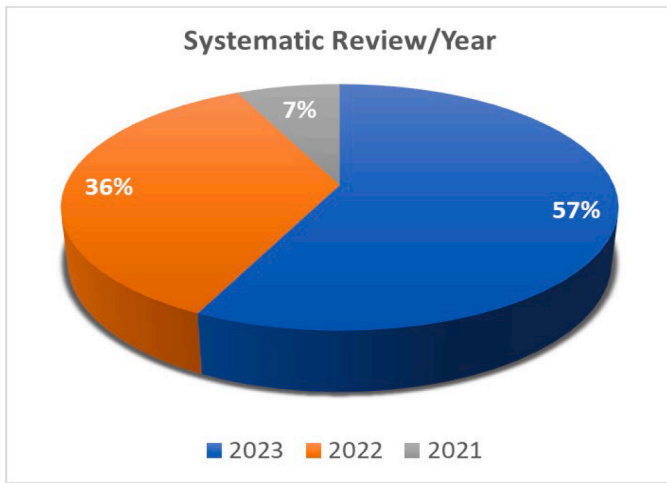


Fig. 8. Number of selected systematic reviews published per year.

study (Alsaleh et al., 2023) suggests that XAI enhances CDS by offering interpretable insights for improved clinical judgments, whereas a study (Vielhaben et al., 2024) shows that it can reduce the cognitive burden and promote fast actions. Reference (Madumal et al., 2020) stresses understanding AI recommendations for patient engagement, while study (Pinto dos Santos & Baeßler, 2018) promotes transparency in CDS systems to foster confidence. Furthermore, studies (Alsaleh et al., 2023; Barredo Arrieta et al., 2020), and (Madumal et al., 2020) investigate XAI's impact on **Electronic Health Records (EHR)** EHR analysis, data privacy, and comorbidity analysis. Study (Alsaleh et al., 2023) demonstrates XAI's ability to find patterns in EHR data, while study (Barredo Arrieta et al., 2020) focuses on predicting patient outcomes and

reference (Madumal et al., 2020) emphasizes improved data management and patient safety. These studies demonstrate XAI's strong impact on clinical decision-making and EHR use, addressing healthcare challenges like drug-drug interactions and comorbidity prediction.

In the **Finance sector**, study (Weber et al., 2023) stands alone but presents a rich exploration into areas such as credit risk assessment, stock price prediction, and anti-money laundering. This indicates an emerging interest and potential for XAI applications in financial analysis and decision-making processes.

Management, particularly in the context of emergency management and global food security, is addressed in study. This study delineates the unique application of XAI in enhancing responses to emergencies, securing food supplies, and improving Intelligent Transportation Systems (ITS), pointing towards the versatility of XAI in addressing global challenges beyond healthcare and finance.

According to studies, XAI has a wide range of applications across multiple fields. For example, research (Albahri et al., 2023) investigates

Table 3

Quality assessment checklist criteria.

#	Question
1	Are research goals clearly specified?
2	Is the scope of the review specific and clear?
3	Is there a clear description of review criteria?
4	Are reviewed articles listed clearly?
5	Is the quality of included studies evaluated?
6	Are findings analyzed to identify trends and gaps?
7	Are limitations of the review process acknowledged?

Table 4 represents the quality assessment results for each of the 14 studies, clearly demonstrating that all articles met the quality assessment criteria. This confirms their appropriateness for inclusion in the subsequent systematic review.

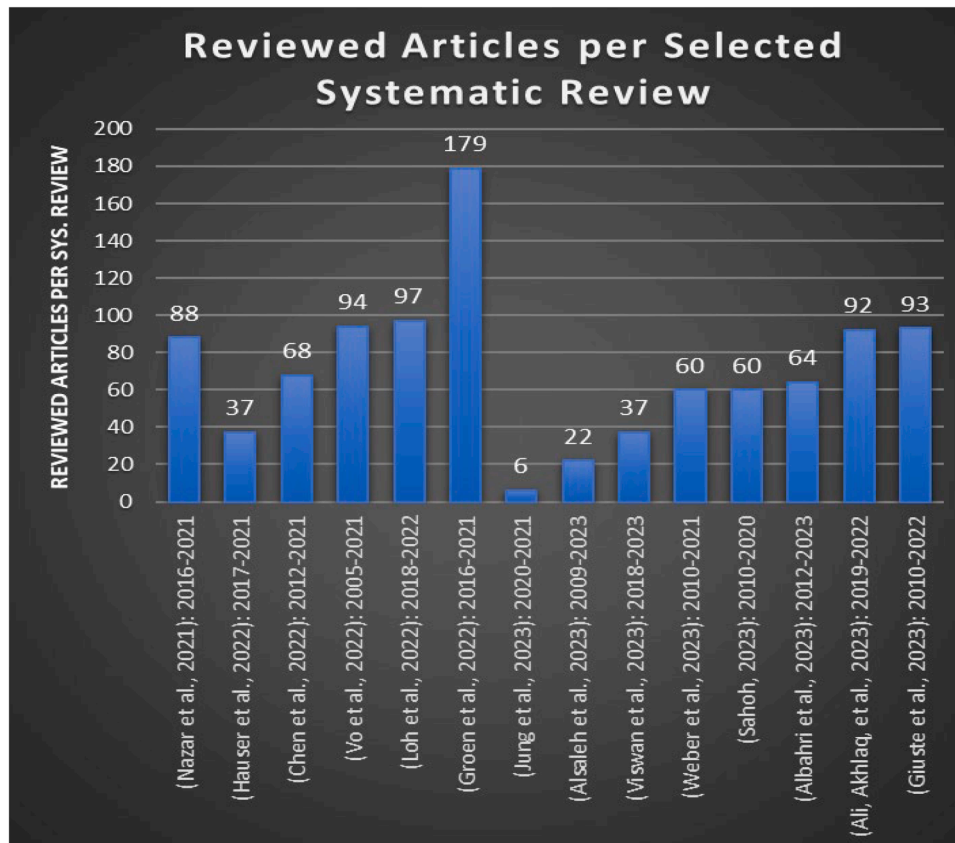


Fig. 9. Number of reviewed articles per selected systematic review.

Table 4

Quality assessment results.

Study	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total	Percentage
S1	1	1	1	0.5	1	1	1	6.5	93 %
S2	1	1	1	1	0.5	1	1	6.5	93 %
S3	1	1	1	0.5	1	1	1	6.5	93 %
S4	1	1	1	1	0.5	1	1	6.5	93 %
S5	1	1	1	0.5	0.5	1	1	6	86 %
S6	1	1	1	1	0.5	1	1	6.5	93 %
S7	1	1	1	1	1	1	1	7	100 %
S8	1	1	0.5	1	0.5	1	1	6	86 %
S9	1	1	1	0.5	0.5	1	1	6	86 %
S10	1	1	1	0.5	0.5	1	1	6	86 %
S11	1	1	1	0.5	0.5	1	1	6	86 %
S12	1	1	1	1	1	1	1	7	100 %
S13	1	1	1	1	0.5	1	1	6.5	93 %
S14	1	1	1	0.5	0.5	1	1	6	86 %

XAI in robotic surgery and human-robot interaction, whereas (Madumal et al., 2020) focuses on 1D biosignal and gait analysis. Ethical considerations like as privacy and justice are discussed in Pinto dos Santos and Baeßler (2018), while (Madumal et al., 2020) and Vo et al. (2022) investigate dermatology and pharmacology, respectively. (Barredo Arrieta et al., 2020) discusses the application of genetic data in healthcare, whereas (Alsaleh et al., 2023) and (Vielhaben et al., 2024) describe XAI's function in public health and epidemic analysis. Furthermore, research (Shin, 2021), and (Groen et al., 2022) look on human-computer interaction, advanced imaging techniques, and aging society issues, respectively.

4.2. RQ2: XAI goals

The analyzed papers reveal a variety of goals aimed at improving the integration and efficacy of AI systems, collectively outlining six major objectives that XAI seeks to achieve, emphasizing the multidimensional character of research and development in this subject. Table 5 outlines the application domains and subdomains of XAI, emphasizing its broad spectrum across sectors such as medical imaging (e.g., diagnostics for glaucoma and Alzheimer's), clinical decision support, electronic health records, disease-specific studies, hospital management, along with applications in finance, logistics, food security, aging society, and intelligent transportation systems. Several research have stressed the need of transparency and trust in AI systems. Study (Albahri et al., 2023) underlines the importance of transparency and suggests approaches to improve explainability, especially in relevant circumstances. A study (Alsaleh et al., 2023) discovered that open disclosure of AI judgments boosts personnel trust in AI. Reference (Groen et al., 2022) studies visual explanations as a technique of boosting user understanding, whereas study (Vielhaben et al., 2024) proposes ethical frameworks for transparency to avoid mistrust. The study (Olivas-Padilla et al., 2024) suggests that simplified explanations promote trust in complex AI models, and the reference (Bai et al., 2021) gives example studies in which increased openness boosts practitioners' reliance on AI. A study (Madumal et al., 2020) underlines the need of transparency in assuring ethical use and accountability in AI, while another study (Pinto dos Santos & Baeßler, 2018) shows how clear, user-friendly explanations foster confidence and acceptance. Finally, reference (Shin, 2021) explores the larger implications of transparency, focusing on its function in establishing confidence across diverse AI applications. Together, these findings demonstrate that increasing transparency is critical to encouraging trust and adoption of AI systems.

Second, seven studies (Albahri et al., 2023; Alsaleh et al., 2023; Bai et al., 2021; Groen et al., 2022; Madumal et al., 2020; Pinto dos Santos & Baeßler, 2018; Shin, 2021) have emphasized the importance of improving decision-making and clinical outcomes. This focus is on using AI to improve decision-making processes, diagnostic accuracy, and personalized medicine. By providing practitioners with understandable

AI-assisted insights, XAI hopes to empower professionals to make better decisions, ultimately improving patient outcomes and increasing overall healthcare delivery efficiency.

The third goal, Increasing Fairness and Reducing Bias, is mentioned in four research (Albahri et al., 2023; Alsaleh et al., 2023; Vielhaben et al., 2024), and (Bai et al., 2021). It emphasizes the need of detecting, reducing, and eliminating biases in AI models in order to promote fair treatment and decision-making for different populations. Fairness is critical for the ethical application of AI technologies, particularly in sensitive domains such as healthcare.

Two studies (Shin, 2021) emphasize the necessity of connecting AI systems with existing legal and ethical frameworks in order to improve regulatory compliance and promote ethical AI use. Ensuring compliance protects patients' rights and increases public trust. These publications argue for continued collaboration among engineers, policymakers, and ethicists to address the issues surrounding AI legislation.

Support for model development and tuning is another topic, as stated in the paper (Olivas-Padilla et al., 2024). This topic covers technical aspects including model visualization, inspection, adjustment, and comparison. The goal is to create models that are both high-performing and interpretable, enabling a better knowledge of AI processes among developers and end users. This emphasizes the need of transparency and interpretability in the iterative process of improving AI models.

Finally, three research (Madumal et al., 2020) underscore the necessity for XAI models to be practical and usable in real clinical situations. XAI strives to meet individual patient demands by enabling tailored treatment programs, hence improving intervention effectiveness. This technique demonstrates AI's potential to transform customized medicine by making it more adaptive and responsive to each patient's specific circumstances. Table 6 summarizes the key goals of XAI as identified in the evaluated studies. It covers major strategic aims such as improving openness and confidence, improving decision-making and clinical results, increasing fairness and decreasing bias, facilitating regulatory compliance, assisting with model creation and tuning, and encouraging real-world application and individualized care. The table also provides particular examples from several research that demonstrate how these aims are pursued across diverse application domains, illustrating the multidimensional character of XAI's goals in developing AI transparency and trustworthiness.

4.3. RQ3: XAI main methods

The reviewed studies offer a comprehensive overview of the main methods employed in XAI. This synthesis, structured around five research questions (RQs), focuses on delineating the primary methods utilized in XAI research, as evidenced by the thematic categorization of the findings.

Visual explanation approaches, such as Layer-wise Relevance Propagation (LRP), Class Activation Maps (CAMs), and Grad-CAM, help to improve AI interpretability, particularly in medical imaging. For example, (Albahri et al., 2023) investigates LRP's capacity to provide insights into deep learning decision-making, whereas (Alsaleh et al., 2023) proposes CAMs as tools for determining the most influential regions in images. Grad-CAM is critical in producing visual explanations that build user trust (Olivas-Padilla et al., 2024), and when combined with heatmaps and saliency maps, these technologies improve transparency in complex datasets, facilitating user understanding in high-stakes contexts such as healthcare.

Interpretable Machine Learning Models, such as Random Forest (RF) classifiers and Explainable Boosting Machines (EBMs), are recognized for their inherent interpretability (Albahri et al., 2023; Madumal et al., 2020), and (Pinto dos Santos & Baeßler, 2018). For example, (Albahri et al., 2023) emphasizes the explicit decision routes of RF classifiers, which make them appropriate for transparency-critical applications such as diagnostics. (Madumal et al., 2020) shows that EBMs can make accurate, interpretable forecasts, which is critical when

Table 5

Application domains and subdomains of explainable artificial intelligence (XAI).

	Medical Imaging Domains	Clinical Decision Support (CDS)	Electronic Health Records (EHR)	Disease-Specific Studies	Hospital Management	Other Sub-Domains	Finance Domains	Global Food Security	Aging Society	Intelligent Transportation Systems (ITS)
S1	Diagnostics (glaucoma, Alzheimer's), Fundus Images			Diagnostics, Disease Prediction, Alzheimer's Disease, glaucoma	ICU Management	Robotic Surgery, Patient Monitoring				
S2	X-ray, CT, MRI, Ultrasound			Respiratory Diseases, COVID-19						
S3	CT, X-ray, MRI, Optical Coherence Tomography	Various Diseases, COVID-19		Breast Cancer, Pneumonia, COVID-19, 1D Biosignal Analysis, Gait Analysis	Hospital Management					
S4		Insights for Healthcare Professionals	EHR Systems							
S5	Skin Cancer, Whole Slide Images			Skin Cancer, Histopathology						
S6				Drug-Drug Interactions Prediction						
S7			Comorbidities, Genetic Information	Comorbidities, Genetic Data						
S8	Radiology, Pathology	Healthcare Professionals' Decision-making	Patient Data Analysis							
S9		Decision-making in Healthcare		COVID-19						
S10						Human-Computer Interaction				
S11	Radiology, Pathology, Dermatology, Ophthalmology, Gastroenterology			Abnormality Detection, Cancer Classification						
S12	Neuroimaging (MRI, PET scans)	Clinical Data Analysis		Alzheimer's Disease						
S13							Credit Risk, Stock Price Prediction, Anti-Money Laundering, Transaction Classification			
S14				Diagnostics	Emergency Response	Patient Outcomes		Sustainable Food Production	Elderly Care	Transportation Management and Safety

stakeholder trust is required. Furthermore, (Pinto dos Santos & Baeßler, 2018) emphasizes the relevance of these models in improving user comprehension, particularly for complicated activities such as multi-modal data fusion. The emphasis on intrinsically interpretable models (Bai et al., 2021) reflects an increasing emphasis on transparency and usability in AI development.

Model-Agnostic Explanation Methods, particularly Local Interpretable Model-Agnostic Explanations (LIME) and Shapley Additive Explanations (SHAP), are extensively discussed (Alsaleh et al., 2023; Madumal et al., 2020), and (Pinto dos Santos & Baeßler, 2018). LIME's ability to approximate complex model predictions with basic, interpretable models makes it ideal for debugging and feature importance analysis (Alsaleh et al., 2023). SHAP, based on cooperative game theory (Madumal et al., 2020), provides consistent and equitable attributions of

feature contributions, improving interpretability across models. Both strategies provide precise, individual-level insights that are particularly useful in high-risk industries such as healthcare and banking (Pinto dos Santos & Baeßler, 2018).

The exploration of **Advanced Data Analysis Techniques and Explainability** reveals a nuanced understanding of XAI's application across different domains. Techniques like Graph Analysis and Blockchain Technology (Albahri et al., 2023), as well as specific methods like Case-Based Reasoning (CBR) and NLP-Based Neural Networks (V, indicate the field's versatility and its alignment with specialized requirements, including user-centered design and empirical evaluations (Shin, 2021).

Finally, the inclusion of Algorithmic and Framework-Specific Methods, as well as Causal and Statistical Methods, demonstrates the

Table 6

Strategic goals of XAI for enhancing AI transparency and trust.

	Improve Transparency and Trust	Enhance Decision-making and Clinical Outcomes	Increase Fairness and Reduce Bias	Facilitate Regulatory Compliance and Ethical AI Use	Support Model Development and Tuning	Promote Real-World Application and Personalized Care
S1	Improve Transparency and Trust; Increase Fairness and Reduce Bias; Facilitate Regulatory Compliance	Enhance Decision-making	Increase Fairness and Reduce Bias	Facilitate Regulatory Compliance		
S2	Model visualization and inspection				Model visualization, selection, and debugging	
S3	Enhance Model Transparency; Improve Trust in AI Predictions	Support Clinical Decision-Making; Guide Model Tuning; Facilitate Model Comparison			Guide Model Tuning; Facilitate Model Comparison	
S4	Improve Explainability for Diverse Stakeholders; Enhance Trust and Transparency	Facilitate Real-World Application	Promote Fairness			Facilitate Real-World Application
S5	Improve Interpretability and Trust	Evaluate XAI's Impact on Human Decision-Making	Address Biases and Improve Model Fidelity			
S6	Understanding AI Model Predictions		Enhancement of Model Fairness		Predictive Sensitivity Testing; Creation of White-Box Models	
S7	Enhance Interpretability and Explainability	Improve Predictive Accuracy; Facilitate Personalized Patient Care				Facilitate Personalized Patient Care
S8	Increase Trust and Transparency	Enhance Diagnostic Accuracy; Improve Patient Outcomes	Promote Fairness and Equity			Improve Patient Outcomes
S9	Increase User Trust and Confidence	Improve Model Performance and Reliability; Facilitate Adoption of AI in Clinical Practice	Mitigate Risks of Data Bias and Ensure Fairness			
S10	Improve Explainability and Transparency; Enhance Trust	Enable Better Decision Making		Facilitate Ethical AI Use		
S11	Facilitate Transparency	Enable Efficient and Accurate Medical Image Analysis				
S12	Enhance Model Transparency and Interpretability; Improve Trust Among Medical Practitioners	Facilitate Patient Understanding; Advance Research in AD				
S13	Enhance Trustworthiness; Improve Informativeness		Ensure Fairness			
S14	Enhanced Explainability and Transparency	Improved Decision-Making in High-Stakes Situations			Integration of Causal Inference	

ongoing investigation of new frontiers in XAI. DeepLIFT, Guided Back-Propagation, and Causal Inference approaches demonstrate the field's commitment to expanding our knowledge of AI models beyond traditional metrics and visual representations. As shown in Table 7, these solutions include a wide range of techniques designed to improve transparency and interpretability. Anchors, for example, provide rule-based explanations for decision-making in legal and financial contexts; LRP backpropagates relevance scores to clarify feature contributions, particularly in image recognition and healthcare; Deep Taylor Decomposition visualizes feature importance in computer vision; and Prediction Difference Analysis evaluates model sensitivity by hiding features.

Furthermore, methods such as SHAP use game theory to quantify feature importance globally and locally, while XGNN explains graph neural networks; CAM and Grad-CAM generate heatmaps to highlight influential image regions; causal models use counterfactuals for policy and healthcare decisions; and techniques such as converting image features into descriptive text, gradient-based attribution, and detecting key features through input perturbations further broaden the interaction. Collectively, these methodologies highlight the breadth of XAI approaches aimed at improving transparency, trust, and decision-making across various domains."

4.4. RQ4: XAI current research gaps

The exploration of gaps within current studies reveals a multifaceted landscape of challenges and unmet needs. These gaps, categorized under nine key themes, provide a comprehensive overview of the areas

requiring further investigation and innovation.

Complexity and Understandability emerged as a primary concern, with studies pointing to the inherent complexity of explanations generated by XAI systems (Albahri et al., 2023) and a notable lack of accommodation for diverse stakeholder perspectives (Madumal et al., 2020). This indicates a pressing need for research aimed at simplifying explanations to enhance their accessibility and relevance to users from varied backgrounds and expertise levels. For example, in healthcare, explanations produced by technologies such as Guided Backpropagation or Grad-CAM, while visually enlightening, may be too sophisticated for physicians who lack technical skills, limiting their practical utility (Chazette & Schneider, 2020).

The theme of **Generalizability and Application** emphasizes limitations in current XAI methods, such as a lack of generalizability across models and domains (Albahri et al., 2023; Pinto dos Santos & Baeßler, 2018; Vielhaben et al., 2024), a narrow focus on specific application areas (Shin, 2021), and difficulties in applying XAI to non-tabular data such as text and images. In healthcare diagnostic systems, for example, approaches such as LIME and SHAP frequently require tailored adjustment to specific models, reducing their usefulness across diverse clinical datasets or machine learning architectures. The study (Albahri et al., 2023) investigates these problems, finding that while LIME and SHAP provide essential insights, their explanations might differ greatly when applied with other models, such as Random Forests versus neural networks, making standardized evaluation difficult. Similarly, (Vielhaben et al., 2024) emphasizes the difficulty of applying typical interpretability approaches to unstructured data such as medical photographs or

Table 7
Comprehensive overview of the main methods employed In XAI.

	Visual Explanation Techniques	Interpretable Machine Learning Models	Model-Agnostic Explanation Methods	Advanced Data Analysis Techniques	Explainability in Specific Contexts	Algorithmic and Framework-Specific Methods	Causal and Statistical Methods
S1	LRP, CAMs	RF Classifiers		Graph Analysis, Blockchain Technology			
S2	CAM, Feature Maps			Dimensionality Reduction (t-SNE, PCA)		Backpropagation Techniques	
S3	SHAP, GradCAM, LIME, Fuzzy Classifiers, Heatmaps	EBM, CBR, Rule-Based Systems	SHAP, LIME				
S4			SHAP, LIME		Visual Analytic Systems		
S5	Heatmaps, Prototype-based explanations	Inherently Interpretable Models			Post-hoc Explanations	Custom XAI Methods	
S6	Gradient-Based Attribution, Guided BackPropagation				NLP-Based Neural Networks	DeepLIFT, Guided BackPropagation	
S7			Feature Importance Estimation	Comorbidity Networks Construction			AUC, Accuracy Metrics
S8	CAM, GradCAM		LIME, SHAP				
S9			LIME, SHAP	Decision Trees, Cross Validation	Feature Importance Ranking		
S10					User-Centered Design		
S11	Attention Mechanisms, Visualization Techniques	Human-Understandable Features			Empirical User Evaluations		
S12	GradCAM, LRP, Saliency Maps		LIME, SHAP	Model-Agnostic Methods, Framework-Specific Explainers		Framework-Specific Explainers (GNNExplainer)	
S13							Theoretical Models, Post-hoc Explainability
S14							Causal Inference, SCM, RCTs, Supervised ML

free-text clinical notes, limiting their general utility. In healthcare diagnostic systems, for example, approaches such as LIME and SHAP frequently require tailored adjustment to specific models, reducing their usefulness across diverse clinical datasets or machine learning architectures. The study (Albahri et al., 2023) investigates these problems, finding that while LIME and SHAP provide essential insights, their explanations might differ greatly when applied with other models, such as Random Forests versus neural networks, making standardized evaluation difficult. Similarly, (Vielhaben et al., 2024) emphasizes the difficulty of applying typical interpretability approaches to unstructured data such as medical photographs or free-text clinical notes, limiting their general utility. In healthcare diagnostic systems, for example, approaches such as LIME and SHAP frequently require tailored adjustment to specific models, reducing their usefulness across diverse clinical datasets or machine learning architectures. The study (Albahri et al., 2023) investigates these problems, finding that while LIME and SHAP provide essential insights, their explanations might differ greatly when applied with other models, such as Random Forests versus neural networks, making standardized evaluation difficult. Similarly, (Vielhaben et al., 2024) emphasizes the difficulty of applying typical interpretability approaches to unstructured data such as medical photographs or free-text clinical notes, limiting their general utility. Concerns about data privacy are especially prevalent in high-stakes domains like banking. Concerns about data privacy are especially prevalent in high-stakes domains like banking. For example, using approaches such as LIME to explain credit scoring models may mistakenly reveal sensitive financial information if explanations are not carefully handled, posing

ethical and regulatory concerns (Pinto dos Santos & Baeßler, 2018). These problems highlight the important necessity for creating privacy-preserving explanation systems, such as federated or differential privacy-based approaches, that can provide relevant explanations while maintaining data confidentiality. Furthermore, the transferability of explanations—that is, their resilience when applied across multiple domains or models—is frequently restricted. According to the study (Pinto dos Santos & Baeßler, 2018), explanations given in one context may not translate effectively to others, particularly when models are retrained or updated, reducing confidence and reliability. In reality, this can result in conflicting explanations for the same decision across datasets or model versions, leading to user distrust. Table 8 outlines these and other significant problems, highlighting areas such as complexity, generalizability, validation, ethical implications, user emphasis, technique variety, data challenges, and model fidelity that necessitate further research and improvement. Addressing these difficulties, such as domain-specific explanation frameworks, enhanced validation protocols, and privacy-preserving approaches, is critical for furthering the robustness, acceptability, and ethical deployment of XAI systems in a variety of domains.

Evaluation and Validation concerns were frequently cited, highlighting a widespread absence of rigorous methods for assessing the effectiveness of XAI approaches (Olivas-Padilla et al., 2024), challenges in validating these solutions in real-world settings (Albahri et al., 2023), and a lack of standardized evaluation metrics (Alsaleh et al., 2023; Madumal et al., 2020). The identified gaps in performance evaluation (Pinto dos Santos & Baeßler, 2018) and the specific mention of

Table 8
Summary of current research gaps in explainable AI (XAI).

Aspect	Current Status/Challenges	Future Directions
Progress	Significant advances across multiple sectors	Continue expanding applications and demonstrating impact
Main Challenges	Complexity, limited generalizability, bias, insufficient evaluation	Develop simpler, human-centered explanations; enhance robustness
Ethical & Fairness	Bias and fairness issues still prevalent	Discover and eliminate biases; promote justice and ethics
Stakeholder Engagement	Limited involvement of diverse users and stakeholders	Incorporate user feedback; employ user-centered design principles
Technique Diversity	Overreliance on narrow explanation methods	Innovate and diversify explanation techniques
Data & Evaluation	Scarcity of datasets; lack of standardized evaluation methods	Expand data resources; establish standardized, domain-specific metrics
Regulatory & Privacy	Need to balance transparency with data privacy	Develop clear guidelines to preserve privacy while ensuring explainability
Application Areas	Underdeveloped in sectors like smart cities, autonomous vehicles	Broaden application to emerging industries

inadequate validation with medical professionals further emphasize the necessity for methodological advancements in XAI evaluation practices. To address these problems, we used a set of performance metrics designed to measure both the prediction accuracy of the models and the quality of the explanations given. The core predictive performance was assessed using accuracy, precision, recall, and the F1-score, as per typical machine learning evaluation techniques (Tao et al., 2024). Furthermore, explanation-specific measures such as the integrity score and explanation consistency were used to assess the interpretability and trustworthiness of the explanations. The fidelity score assesses how well the explanation mirrors the model's real decision-making process, assuring the explanation's consistency. Explanation consistency measures the stability of explanations under modest input disturbances, which indicates resilience and reliability (T et al., 2022). Our selection of these indicators was motivated by the need to balance traditional model performance metrics with explanation-specific criteria, which are especially important in high-risk sectors such as healthcare, where trust and transparency are essential. This complete evaluation framework seeks to give a more robust and relevant assessment of XAI techniques, hence driving future methodological advancements (T et al., 2022; Tao et al., 2024).

Ethical and Privacy Considerations are also of paramount importance, with studies indicating insufficient attention to ethical implications of XAI systems (Albahri et al., 2023) and concerns regarding data privacy and accessibility (Alsaleh et al., 2023). These gaps call for a heightened focus on the ethical dimensions of explainability and the development of frameworks that prioritize user privacy and data protection.

In terms of **End-User Focus and Usability**, the reviewed literature points to a limited emphasis on aligning XAI outputs with end-user goals (Olivas-Padilla et al., 2024), a recurring lack of diverse stakeholder involvement (Madumal et al., 2020), and a knowledge imbalance that overlooks the principles of user-centered design (Groen et al., 2022). Addressing these gaps is essential for enhancing the usability and accessibility of XAI systems, ensuring that they meet the needs and expectations of their intended users.

Technique Diversity and Innovation highlight the field's overreliance on a narrow range of techniques (Olivas-Padilla et al., 2024) and the limited explainability afforded by existing models (Groen et al., 2022). This suggests a gap in the diversification of approaches and underscores the need for innovative research efforts to expand the toolkit of explainability techniques.

Data Challenges are evident in the scarcity of public datasets for specific domains (Pinto dos Santos & Baeßler, 2018), issues with clinical data quality and availability (Vielhaben et al., 2024), and the lack of comprehensive and balanced datasets. These challenges impede the development and testing of XAI systems, highlighting the need for concerted efforts to improve data resources for XAI research.

Model Fidelity and Accuracy issues, such as the difficulties associated with ensuring model fidelity and the trade-offs between explainability and accuracy (Shin, 2021), point to inherent tensions in XAI development. These concerns necessitate ongoing research to balance the dual objectives of transparency and performance.

Lastly, the study (Alsaleh et al., 2023) underlines the importance of a larger research strategy to address domain-specific issues. This approach is critical for addressing the difficulty of explainability in AI, ensuring that XAI solutions are resilient, versatile, and fulfill a wide range of user needs across industries. Table 9 identifies the major gaps in present XAI research “

4.5. RQ5: XAI research recommendations

The systematic evaluation of 14 papers in the field of XAI offers crucial recommendations for furthering the discipline, concentrating on common themes that stress the necessity for a multidimensional approach. These themes address the problems of making AI systems more explainable and transparent.

The necessity for strong AI frameworks and standards is a common issue throughout the studies. Studies [(Albahri et al., 2023; Alsaleh et al., 2023), and (Madumal et al., 2020)] emphasize the relevance of regulatory and ethical frameworks in addressing AI's complex ethical challenges. Standardized techniques for data processing and model validation, as mentioned in Barredo Arrieta et al. (2020), are critical for ensuring accountability and transparency in XAI. Furthermore, research (Madumal et al., 2020) and (Shin, 2021) propose for complete frameworks to provide ethical, effective AI explanations, allowing for a consistent assessment of XAI applications.

Enhancing explainability and transparency is a major focus in studies (Albahri et al., 2023; Bai et al., 2021; Barredo Arrieta et al., 2020; Madumal et al., 2020; Vielhaben et al., 2024), and (Pinto dos Santos & Baeßler, 2018), stressing their importance for ethical AI deployment. Albahri et al. (Albahri et al., 2023) emphasize transparency in healthcare AI to increase user confidence, whereas another study (Barredo Arrieta et al., 2020) pushes for interpretable models to simplify complicated systems. Study (Vielhaben et al., 2024) emphasizes the need of explainability in high-stakes scenarios, while (Bai et al., 2021) focuses on model consistency for credible explanations. Study (Madumal et al., 2020) focuses on legal frameworks for accountability, whereas (Pinto dos Santos & Baeßler, 2018) investigates how clear explanations might enhance user adoption. These studies try to make AI systems more understandable, hence increasing trust and acceptance.

Several research (Albahri et al., 2023; Alsaleh et al., 2023; Barredo Arrieta et al., 2020; Pinto dos Santos & Baeßler, 2018; Selvaraju et al., 2017), and (Soori et al., 2023) emphasize the **Significance of Promoting Collaboration and Multidisciplinary Integration in XAI**. Albahri et al. (Albahri et al., 2023) advocate for multidisciplinary approaches in healthcare AI, whereas (Alsaleh et al., 2023) emphasizes the importance of ethical considerations in AI research. The study (Barredo Arrieta et al., 2020) stresses collaboration between technical and non-technical parties to increase AI's societal relevance. Furthermore, (Pinto dos Santos & Baeßler, 2018) investigates the importance of multiple viewpoints in building XAI solutions, while (Selvaraju et al., 2017) and (Soori et al., 2023) emphasize the role of politicians and ethicists in ensuring XAI meets the interests and values of many groups. This collaboration is vital for developing socially meaningful, morally sound XAI systems.

Prioritizing Societal and Clinical Needs is a common theme in studies (Alsaleh et al., 2023; Madumal et al., 2020; Olivas-Padilla et al.,

Table 9
XAI research gaps.

XAI method	Complexity of Explanations	Lack of Generalizability	Limited Validation in Real-World Settings	Insufficient Emphasis on Ethical Considerations			
S2			Lack of XAI Effectiveness Evaluation, Limited Focus on End-User Goals	Limited Focus on End-User Goals	Overreliance on Few Techniques		
S3		Limited Focus on 1D Biosignals and Clinical Notes	Challenges in Performance Evaluation			Lack of Public Datasets for Certain Domains	
S4	Lack of Diverse Stakeholder Accommodation		Uncertainty in Clinical Deployment, Absence of Standardized Evaluation Methods	Lack of Diverse Stakeholder Accommodation			
S5						Challenges with Model Fidelity and Post-hoc Explanations	
S6		Limited Application in Non-Tabular Data					
S7			Lack of Standard Evaluation Method for XAI		Limited Explainability of Models		
S8		Domain-Specific Challenges	Lack of Standardization, Data Privacy and Accessibility	Data Privacy and Accessibility			Domain-Specific Challenges
S9		Difficulty in Ensuring Model Generalizability				Challenges in Clinical Data Quality and Availability	
S10		Limited Scope of Study, Concentration on Specific Areas					Challenges in Explainability and Model Accuracy
S11				Limited Access to End Users, Knowledge Imbalance, Neglect of User-Centered Design Principles			
S12		Limited Comprehensive and Balanced Datasets, Unaddressed Multimodal Data Interpretations	Lack of Validation with Medical Professionals, Absence of Ground Truth for Explainability				
S13		Limited Application Areas					Neglect of Theory-Building, Lack of Regulatory Compliance Discussion
S14		Need for More Research Contributions Across Various Fields					Insufficient Integration of Association, Intervention, and Counterfactual Levels, Limited Practical Implementations of Counterfactual Reasoning

2024), and (Pinto dos Santos & Baeßler, 2018), indicating an increasing recognition of the importance of grounding AI research in real-world contexts. Study (Alsaleh et al., 2023) emphasizes the importance of adapting AI solutions to solve public health challenges. The study (Olivas-Padilla et al., 2024) focuses on providing XAI solutions to specific therapeutic situations. Research (Madumal et al., 2020) examines the significance of regulatory frameworks in ensuring that AI technologies satisfy social objectives, whereas study (Pinto dos Santos & Baeßler, 2018) investigates how adapting XAI to real-world situations improves its effectiveness. Overall, our findings highlight the significance of developing XAI solutions that are both technically possible and helpful to society and healthcare.

The urge to increase the scope of **Advancing Research and Innovation** within the XAI sector is stressed in research (Olivas-Padilla et al., 2024; Pinto dos Santos & Baeßler, 2018; Sahoh & Choksuriwong, 2023; Shin, 2021), indicating a desire to push the boundaries. Study (Olivas-Padilla et al., 2024) supports investigating underrepresented methodologies in XAI, whereas (Pinto dos Santos & Baeßler, 2018) emphasizes the necessity of continual innovation to improve AI applications. Research (Shin, 2021) stresses the importance of multidisciplinary approaches in enriching the area, whereas (Sahoh & Choksuriwong, 2023) emphasizes the study of new fields and applications. This pursuit of varied XAI methodologies not only enriches the research environment but also offers up new avenues for effectively tackling complex ethical concerns, establishing a lively community dedicated to advancing the state of the art in XAI.

The importance of **Ethical Considerations**, fairness, and data privacy is highlighted in research (Albahri et al., 2023; Alsaleh et al., 2023; Madumal et al., 2020), emphasizing the crucial necessity for responsible AI development. (Albahri et al., 2023) emphasize the need of openness and ethical integrity in healthcare AI for establishing user confidence. The study (Alsaleh et al., 2023) focuses on the importance of introducing fairness into AI algorithms to avoid prejudice and assure equitable results. Furthermore, research (Madumal et al., 2020) highlights the need of data privacy in AI systems, calling for frameworks that secure sensitive information while being transparent. Together, these studies

demonstrate a fundamental commitment to developing AI technologies that adhere to ethical standards and emphasize human rights. **Human-Centered Design** principles (Groen et al., 2022) reflects a foundational commitment to ensuring that XAI technologies are developed and deployed in ways that uphold the highest standards of ethical integrity and human values. Incorporating clinical expertise (Barredo Arrieta et al., 2020) and prioritizing human-centric design approaches ensure that XAI systems are not only technically proficient but also ethically aligned and practically useful, thereby reinforcing the importance of ethical considerations and human values in guiding the development and application of AI technologies. Table 10 presents the recommendations in XAI research, emphasizing the importance of advancing explanation techniques, improving model transparency and interpretability, fostering interdisciplinary collaboration, ensuring ethical and responsible AI deployment, enhancing user-centered design, and promoting validation and benchmarking efforts. These recommendations aim to address current gaps, improve practical applicability, and facilitate the development of trustworthy and effective XAI systems across diverse fields.

5. Discussion

The exploration of XAI across various domains highlights its goals, methodologies, gaps, and future research recommendations. This analysis synthesizes findings from 14 systematic reviews published between 2021 and 2023, organized around five key research questions. It provides a comprehensive overview of the current state and future directions of XAI research.

5.1. RQ1 overview of XAI applications across domains

The analysis identifies three principal domains where XAI is employed, with healthcare accounting for 86 % (12 out of 14 studies) and 87.3 % of the total reviewed publications, or 810 out of 928 (excluding replications). This is followed by contributions in finance and management, which account for 7 % (one study for finance and one for

Table 10
XAI research recommendations.

	Developing Frameworks and Standards	Enhancing Explainability and Transparency	Fostering Collaboration and Multidisciplinary Integration	Prioritizing Societal and Clinical Needs	Advancing Research and Innovation	Ethical Considerations and Human-Centered Design
S1	Regulatory and Ethical Frameworks	Model Explainability and Transparency	Multi-Stakeholder Collaboration			Ethical Guidelines and Fairness
S2	Evaluation Methods			Addressing Clinical and Societal Needs	Broad Research and Innovation	
S3		Model Explainability and Transparency	Multi-Stakeholder Collaboration	Addressing Clinical and Societal Needs	Broad Research and Innovation	
S4	Regulatory and Ethical Frameworks, Comprehensive Frameworks	Model Explainability and Transparency		Addressing Clinical and Societal Needs		Ethical Guidelines and Fairness
S5		Model Fidelity				
S6				Addressing Clinical and Societal Needs	Broad Research and Innovation	Ethical Guidelines and Fairness
S7	Data Collection and Model Validation	Transparent and Interpretable Models	Multi-Stakeholder Collaboration, Incorporating Clinical Expertise			
S8	Regulatory and Ethical Frameworks		Multi-Stakeholder Collaboration	Addressing Clinical and Societal Needs		Ethical Guidelines and Fairness
S9		Model Explainability and Transparency				
S10	Comprehensive Frameworks				Broad Research and Innovation	
S11						Human-Centered Design
S12						Incorporating Clinical Expertise
S13					Expanding Research to Underexplored Areas	
S14			Interdisciplinary Collaboration		Broad Research and Innovation	

management) and 6.35 % of the total evaluated articles, respectively, resulting in 59 articles for finance and management out of 928 (excluding replications).

5.1.1. Healthcare: A front-runner in XAI application

This domain is a crucial area for XAI use, emphasizing the importance of openness, accuracy, and explainability in diagnosis and treatment planning. Studies reveal that XAI has a significant impact in diagnostics, disease prediction, medical imaging, clinical decision support, electronic health records, and disease-specific research, solving the field's numerous issues (Shin, 2021).

Enhancing Diagnostics and Treatment: The research highlights XAI's transformative potential in diagnostics, disease prediction, medical imaging, clinical decision support, and electronic health records. It enhances illness detection and tracking, especially for conditions like Alzheimer's and glaucoma, and improves the interpretation of complex imaging data, leading to quicker and more accurate diagnoses for diseases such as COVID-19 and breast cancer.

Augmenting Clinical Decision-Making: The application of XAI in CDS and EHR analysis indicates a move towards personalized medicine, where XAI supports healthcare professionals by providing actionable insights derived from vast data sets, enhancing patient care and outcomes.

5.1.2. Finance: emerging interest and potential

The study by Soori et al. (2023) sheds light on the burgeoning interest in applying XAI within the finance sector, exploring critical areas such as credit risk assessment, stock price prediction, and anti-money laundering. This solitary but significant contribution suggests a nascent yet promising area for XAI application, potentially revolutionizing how financial data is analyzed and decisions are made.

Revolutionizing Financial Analysis: XAI's ability to provide insights can enhance trust and reliability in automated financial decision-making processes, offering a pathway to more transparent and accountable financial services.

5.1.3. Management: addressing global challenges

The unique application of XAI in management, particularly in emergency management and global food security, as discussed by Shin (2021), points to the versatility of XAI beyond traditional domains.

Enhancing Emergency Response and Global Food Security: XAI's application in intelligent transportation systems (ITS), emergency management, and food security underscores its potential to contribute to solving critical global challenges, improving response strategies, and ensuring sustainable practices.

5.2. RQ2: XAI goals

The results for RQ2 provide a comprehensive overview of the current XAI goals, with a notable focus transparency, enhancing decision-making and reducing bias.

5.2.1. Improving transparency and trust

Transparency and trust are required for integrating AI into key sectors such as healthcare, finance, and management. Studies emphasize the importance of AI systems being intelligible and reliable to users, ensuring that AI choices are clear and trustworthy. This improves AI adoption by demystifying the decision-making process, resulting in increased acceptance and reliance on AI solutions.

5.2.2. Enhancement of decision-making and clinical outcomes

The emphasis on improving decision-making and clinical outcomes illustrates XAI's ability to provide better patient care. XAI helps healthcare personnel make better decisions by delivering clear, interpretable data. This is consistent with the concept of customized medicine, in which therapies are adapted to specific patient needs, resulting

in more successful therapy.

5.2.3. Increasing fairness and reducing bias

Ensuring fairness and eliminating bias in AI models is critical for providing equitable treatment to patients with various demographics. This ethical imperative strives to create AI systems that are free of intrinsic bias, particularly in healthcare, where biased models can have a substantial impact on patient treatment and results.

5.2.4. Facilitate regulatory compliance and ethical AI use

The emphasis on regulatory compliance and ethical AI use highlights the importance of aligning AI with legal and ethical standards. This protects patients' rights and privacy while encouraging responsible AI development. Ensuring compliance improves both individual safety and social acceptability of AI.

5.2.5. Support model development and tuning

Supporting model creation and refinement is critical to enhancing XAI. This entails improving model visibility, inspection, and comparison, enabling developers to construct AI systems that are both performant and interpretable. This approach guarantees that AI models remain effective and intelligible, hence increasing their value and application in real-world circumstances.

5.2.6. Promote real-world application and personalized care

Promoting real-world applicability and personalized care demonstrates XAI's goal of making technology usable in routine clinical settings. XAI improves personalization by developing AI models that are suited to patient demands, hence boosting the efficacy of interventions and results. This highlights AI's revolutionary power in creating a more adaptable, responsive, and patient-centered system.

5.3. RQ3: XAI main methods

The contribution of this study includes the classification of XAI methods (+30) into seven groups, elucidating their significance, applications, and implications for the advancement of XAI.

Group 1: Visual Explanation Techniques

Visual explanation approaches such as Layer-wise Relevance Propagation (LRP), Class Activation Maps (CAMs), and GradCAM are critical for understanding how deep learning models work. These approaches are extremely useful in industries such as medicine and neuroimaging because they reveal crucial locations within images or signals. By demonstrating how AI models process and prioritize information, these strategies improve user understanding and confidence.

Group 2: Interpretable Machine Learning Models

Interpretable Machine Learning Models like Random Forest (RF) Classifiers and Explainable Boosting Machines (EBM) emphasize the value of inherent interpretability. These models are particularly beneficial in scenarios where clear, understandable outcomes are essential, facilitating their application in diverse tasks ranging from multimodal data fusion to precise predictions. The preference for inherently interpretable models highlights a shift towards designing AI systems that prioritize understandability from the outset.

Group3: Model-Agnostic Explanation Methods

Model-Agnostic Explanation Methods, such as Local Interpretable Model Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP), are crucial for dissecting the contributions of individual features to model predictions. These techniques offer detailed insights on a per-instance basis, proving invaluable for tailored analysis and

interpretation across various applications. Their versatility allows for broad application across different AI models, enhancing the granularity and specificity of explanations.

Group 4: Advanced Data Analysis Techniques

The inclusion of Advanced Data Analysis Techniques, such as graph analysis and blockchain technology, alongside specific methods like Case-Based Reasoning (CBR) and NLP-Based Neural Networks, showcases XAI's adaptability and alignment with specialized requirements. These techniques facilitate a nuanced understanding of data and model behavior, catering to unique demands such as user-centered design and empirical evaluations. This category underscores the field's innovation and its commitment to addressing complex analytical challenges.

Group 5: Explainability in Specific Contexts

This category underscores the tailored application of XAI methods to meet the unique requirements of specific domains or contexts, such as NLP-based neural networks for language processing or custom XAI methods designed for challenges. It highlights the importance of developing and applying explainability techniques that are not just general but also finely tuned to the nuances of individual application areas.

Group 6: Algorithmic and Framework-Specific Methods

Exploring new frontiers in XAI, Algorithmic and Framework-Specific Methods include techniques like DeepLIFT and Guided Back-Propagation. These methods extend beyond traditional visual explanations, offering deeper insights into model behavior and decision-making processes. Their development reflects the ongoing effort to enhance our understanding of AI models, pushing the boundaries of what can be achieved in terms of explainability and interpretability.

Group 7: Causal and Statistical Methods

Causal and Statistical Methods, including Causal Inference techniques and Statistical Model Checking (SMC), represent an advanced approach to understanding the underlying mechanisms and relationships within AI models. These methods aim to provide a more profound understanding of causality and influence, offering a robust framework for interpreting AI decisions in a context that mirrors real-world complexities.

5.4. RQ4: XAI current research gaps

The identification of gaps within current XAI research highlights areas requiring further investigation and innovation. These gaps encompass challenges related to complexity and understandability, generalizability and application, evaluation and validation, ethical and privacy considerations, end-user focus and usability, technique diversity and innovation, data challenges, model fidelity and accuracy.

5.4.1. Complexity and understandability

The challenge of complexity and understandability underscores a fundamental issue in XAI: making explanations generated by AI systems accessible and relevant to users with varied backgrounds. Simplifying explanations without compromising their accuracy or depth is essential for enhancing user engagement and trust in AI systems. Future research should focus on developing methods that can adapt explanations based on the user's expertise level, promoting wider adoption and understanding.

5.4.2. Generalizability and application

The limited generalizability and narrow focus of current XAI

applications highlight a significant gap in the field's ability to cater to diverse models and domains. Expanding the adaptability and applicability of XAI solutions across different data types, including non-tabular data and natural language, is critical. Research efforts should aim to develop more versatile XAI frameworks that can seamlessly integrate with various AI models and applications.

5.4.3. Evaluation and validation

The absence of rigorous, standardized methods for assessing XAI effectiveness points to a crucial need for establishing clear evaluation and validation frameworks. This includes the development of benchmarks and metrics that can objectively measure the quality of explanations and their impact on user decisions in real-world settings. Engaging with domain experts, particularly in sensitive areas like healthcare, can provide valuable insights into the practical utility of XAI solutions.

5.4.4. Ethical and privacy considerations

Addressing ethical and privacy concerns is vital since XAI systems can influence decision-making in critical industries. Future research should prioritize developing ethical principles and privacy-preserving approaches for XAI design and implementation. This includes assuring data transparency, protecting user data, and minimizing biases in AI models.

5.4.5. End-user focus and usability

The disparity in end-user focuses and usability emphasizes the necessity to connect XAI outputs with user demands and goals. Adopting user-centered design concepts and integrating a diverse range of stakeholders can help XAI systems become more usable and accessible. Research should also investigate interactive, user-friendly interfaces to improve comprehension of AI explanations.

5.4.6. Technique diversity and innovation

The overreliance on a narrow range of techniques and the limited explainability of current models call for a diversification of approaches in XAI research. Innovating beyond conventional methods and exploring new paradigms in explainability can broaden the field's capabilities. This includes leveraging advancements in computational techniques and exploring interdisciplinary approaches to develop novel XAI models.

5.4.7. Data challenges

The absence of available datasets, data quality difficulties, and the necessity for comprehensive, balanced datasets all pose significant challenges to XAI development. Collecting, curating, and distributing high-quality datasets targeted to XAI research is critical. Collaboration across academia, industry, and government can improve access to diverse and representative datasets, hence enabling more rigorous XAI research.

To solve data restrictions, future research should focus on collaborative dataset curation across fields. Strategies include setting labeling standards for fairness and uniformity, enhancing minority classes for balance, and including synthetic data generation in areas where real-world datasets are rare. These activities are critical to guaranteeing the reliability and ethical integrity of XAI systems.

5.4.8. Model fidelity and accuracy

Balancing model fidelity with the need for explainability presents an inherent tension in XAI development. Research must continue to explore methods that maintain high accuracy while providing meaningful explanations. Investigating trade-offs and synergies between explainability and performance can yield insights into optimizing both aspects in AI models.

5.4.9. Interdisciplinary and cross-domain research

The call for interdisciplinary and cross-domain research reflects the

complex, multifaceted nature of explainability in AI. Bridging gaps between different fields, including computer science, cognitive science, ethics, and domain-specific knowledge, is crucial for developing comprehensive XAI solutions. Collaborative research efforts can address domain-specific challenges and integrate diverse perspectives, enriching the XAI landscape.

5.5. RQ5: XAI research recommendations

The recommendations derived from the systematic review emphasize the need for establishing robust frameworks and standards, enhancing explainability and transparency, fostering collaboration and multidisciplinary integration, prioritizing societal and clinical needs, advancing research and innovation, and incorporating ethical considerations and human-centered design principles.

5.5.1. Establishment of robust frameworks and standards

Studies underline the importance of strong frameworks and standards in AI research, which include regulatory and ethical guidelines as well as approaches for data processing, model validation, and evaluation. These frameworks are critical for resolving ethical concerns in AI and enabling accountability and transparency in XAI systems. Comprehensive frameworks will allow for consistent assessments, increasing trust and reliability in XAI applications.

5.5.2. Enhancing explainability and transparency

The research emphasizes the need to improve explainability and transparency in AI systems by building mechanisms that provide clear, intelligible explanations for AI judgments. The emphasis on model quality and the creation of interpretable models are intended to make AI complexity more accessible. Improving explainability is both practical and ethical, as it is critical for establishing user confidence and assuring ethical use of AI technologies.

5.5.3. Fostering collaboration and multidisciplinary integration

The recommendations emphasize the value of collaboration and multidisciplinary integration in addressing XAI concerns that go beyond technological issues. Integrating insights from healthcare, policy-making, ethics, and other disciplines is critical for developing XAI systems that are technically solid, socially relevant, and ethically grounded. Collaborative efforts will assist guarantee that XAI technologies suit the diverse requirements and values of different communities, thereby increasing their social effect.

5.5.4. Prioritizing societal and clinical needs

Tailoring XAI to specific societal and clinical needs is critical for assuring its impact. This emphasis on practical applicability emphasizes the significance of developing XAI solutions that are not just creative but also socially beneficial and therapeutically effective. By basing research in real-world contexts, XAI technologies can successfully meet current social requirements, enhancing their potential to improve society well-being.

5.5.5. Advancing research and innovation

There is a strong desire to broaden research and innovation in XAI by investigating underrepresented methodologies and pushing the limits of existing knowledge. This expansion into new fields, applications, and approaches improves the profession while also providing chances to handle complicated challenges. Such a passion for invention creates a vibrant research community dedicated to enhancing XAI. Finally, it seeks to improve the efficacy and application of XAI solutions.

5.5.6. Ethical considerations and human-centered design

The emphasis on ethical issues, fairness, data protection, and human-centered design in XAI development demonstrates a dedication to upholding high ethical standards. By incorporating clinical experience, XAI

solutions become both technically sophisticated and ethically sound. This approach emphasizes the importance of ethical and human values in AI development.

5.5.7. Practical implications and case studies

To deepen the analysis and provide practical insights into the application of XAI, we propose incorporating detailed case studies and examples. These case studies show how XAI approaches are employed in various fields, the challenges encountered, and the solutions discovered (Table 11). This systematic review examined current trends in XAI research, with a special emphasis on healthcare applications. As shown in Table 11, the distribution of research across domains reveals a strong emphasis on medical imaging, which is consistent with recent advances in diagnostic technologies.

5.5.8. Implementation and computational feasibility of XAI methods

XAI approaches differ in their computing requirements and simplicity of integration. Although commonly used, methods such as SHAP and LIME can be computationally expensive, rendering them unsuitable for real-time systems. Grad-CAM and Integrated Gradients provide speedier explanations, but they require access to model internals. Explanations in fields such as healthcare and finance must not only be understandable, but also efficient and compatible with regulatory requirements. Model accuracy and explanation transparency are frequently at odds, influencing deployment decisions.

Overall, successful XAI implementation necessitates balancing interpretability, computational cost, and domain-specific limitations to allow practical and scalable use.

5.5.9. Distinctiveness and practical significance of this systematic review

This review is distinguished from previous XAI literature by its distinct methodological and scope-related characteristics. Unlike prior narrative or targeted reviews, we conducted a systematic review of systematic reviews while carefully adhering to PRISMA principles. We conducted a comprehensive analysis of 997 publications, which were reduced to 328 high-quality reviews published in Q1 journals, assuring a focus on robust, peer-reviewed sources.

Table 11
XAI research case studied.

Domain	Case Study	Examples	Outcome
Healthcare	Implementing XAI in diagnostic systems to identify diseases from medical images.	Detailed analysis of a machine learning model used to detect diabetic retinopathy, showcasing the application of explainable methods to ensure transparency and trust among medical professionals.	Improvements in diagnostic accuracy, clinician trust, and patient outcomes.
Finance	Utilizing XAI in credit scoring models to enhance decision-making.	Examination of XAI techniques in interpreting credit risk assessments, ensuring fairness and regulatory compliance.	Enhancements in customer trust and regulatory adherence.
Management	Adoption of XAI in decision support systems for business management.	Analysis of an XAI-powered tool aiding strategic decision-making by providing interpretable insights from complex data sets.	Better decision quality and stakeholder confidence.

Our multi-domain coverage includes not only well-known industries like healthcare and banking, but also emerging fields such as emergency management and food security. This broad scope allows us to synthesize diverse applications and identify overarching challenges and opportunities in XAI, providing a comprehensive landscape that many prior reviews lack.

Furthermore, our emphasis on high-impact, peer-reviewed publications strengthens the credibility and relevance of our findings, making them more useful to both academia and industry. Our evaluation provides a platform for future research and practical application by incorporating the most recent accomplishments across fields.

6. Conclusion

This systematic review focuses on the considerable advances made in the field of explainable artificial intelligence (XAI), illustrating its growing applicability in a variety of sectors including healthcare, finance, autonomous systems, and smart cities. It emphasizes the importance of XAI in creating trust, transparency, and ethical use of AI technologies. This review's primary strength is its meta-level synthesis of multiple scholarly publications, which provides a thorough perspective that sets it apart from the field's traditional literature.

Despite these advances, some ongoing issues exist. These include the inherent complexity of explanations, the limitations of cross-domain generalizability, and the requirement for rigorous, standardized evaluation techniques.

Furthermore, issues of bias, fairness, and stakeholder participation are not well addressed, impeding the development of accessible, egalitarian, and user-centric XAI systems—particularly in high-risk industries such as healthcare, banking, and autonomous cars.

The assessment also points out major weaknesses, such as an overreliance on narrow explanation tactics and insufficient attention to ethical problems. To speed development, future research should focus on providing straightforward, human-centered explanations in conjunction with cognitive scientists, enhancing methodology diversity, and expanding data resources for underrepresented domains. Furthermore, building comprehensive review procedures and regulatory requirements is critical for assuring responsible, trustworthy, and privacy-preserving AI systems. Moving forward, efforts should concentrate on simplifying explanation approaches for non-expert users, widening XAI applications in emergent and underexplored sectors, and developing standardized frameworks for review and regulation. Through targeted research, interdisciplinary collaboration, and policy development, XAI has the ability to fully realize its promise of providing ethical, transparent, and socially beneficial AI systems that are both accessible and valuable to society as a whole.

7. Future work

Several practical recommendations are made to help guide future XAI research. These include creating new explanation approaches that are simple enough for non-experts to understand, with help from cognitive scientists. Researchers should also work to improve justice by discovering and eliminating biases in AI models, particularly in sensitive fields such as healthcare and finance. Improving accessibility is critical, making XAI tools more usable for practitioners of all skill levels. Furthermore, it is critical to develop XAI into emerging industries such as self-driving vehicles and smart cities. Finally, maintaining a balance between transparency and privacy is critical, with guidelines required to preserve sensitive data while giving valuable explanations. These activities will overcome present limits and stimulate innovation in XAI.

Declaration of competing interest

The authors declare that there are no conflicts of interest regarding the publication of this paper. This research was conducted

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Supplementary materials

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Data availability

I uploaded the dataset files.

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